Department of Electrical, Electronic and Computer Engineering

Assessment ID: 2021EBN111A09

Electricity and Electronics EBN 111

Release date: 25 June 2021

Marks:	20	Total number of pages:	8
Submission Deadline:	29 June 2021 (23:59)	Open/Closed book:	Open

Important instructions

1. This PDF file must be used in together with the **Excel answer spreadsheet** found on your AMS system account. The **Excel answer spreadsheet** offers each of you the unique parameter values for each question. Some extra instructions are also available on the **Excel answer spreadsheet**.
2. Final answers must be written as real numbers and not as fractions. (Use at least 5 significant figures to write irrational numbers.)
3. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the **Excel answer spreadsheet**.)
4. You can upload the **Excel answer spreadsheet** to AMS multiple times before the submission deadline. You can self-grade two attempts of these uploads to check and possibly improve your answers. The mark for each assignment is awarded according to the final submission of the **Excel answer spreadsheet**.

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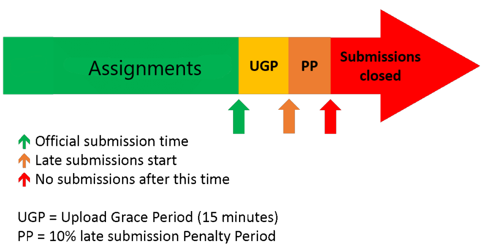
Assistant Lecturers: M. Maritz, J. Masela, J. Thiselton, M. Trivedi, D. Wepener

Online assessment submission rules and time-line

The following rules and time-line apply for all Assignments for EBN111/122 as illustrated by the figure below:

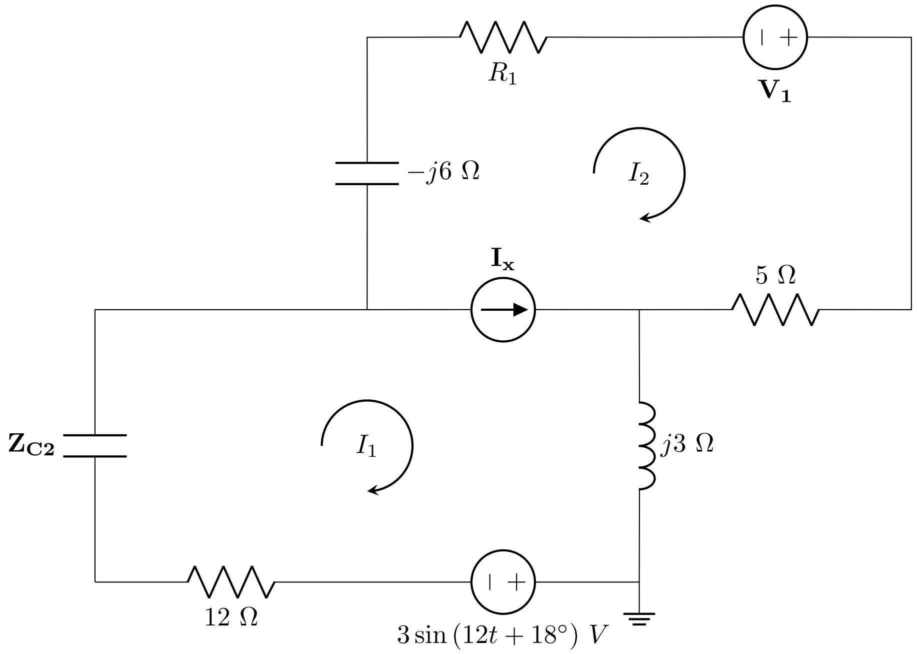
1. The start of the green section in the figure indicates the release of an assignment. The release of the assignment will be announced on ClickUP.
2. The official due date for the assignment is indicated by the green arrow in the figure. The due dates are also shown on the Lecture Schedule on ClickUP and on the AMS.
3. You must submit your **Excel answer spreadsheet** with your answers to each assignment on the AMS before the due date (green arrow in the figure below).
4. If you experience a technical issue with respect to uploading your assignment on the AMS, you can request assistance via **e-mail** ebn2021.ams@gmail.com. Note, except for extraordinary cases, you cannot submit your answers via email.
5. The submission due date will be followed by an Upload Grace Period of 15 minutes during which you must submit your answers on the AMS even if not finished with the assignment. No email submissions will be accepted during this period.
6. The Upload Grace Period is followed by a Penalty Period of 10 minutes. If you submit during this period, a 10% penalty will be applied to your final mark.
7. No late submissions will be accepted after the Penalty Period.

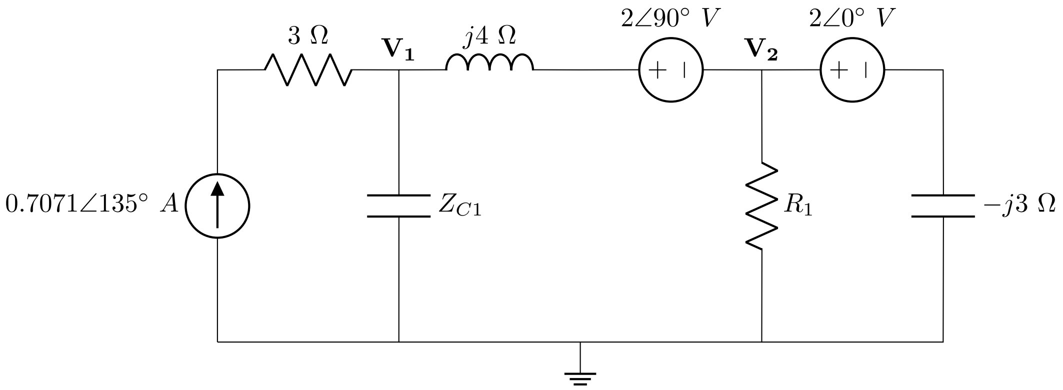


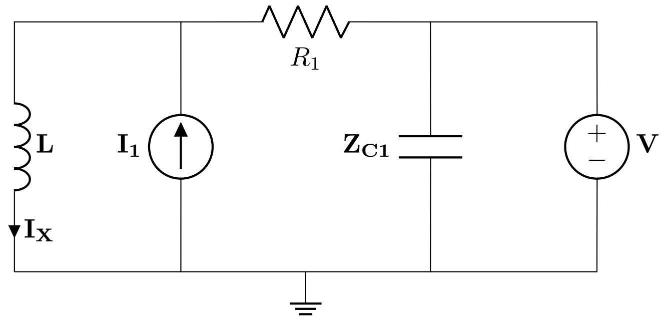
Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

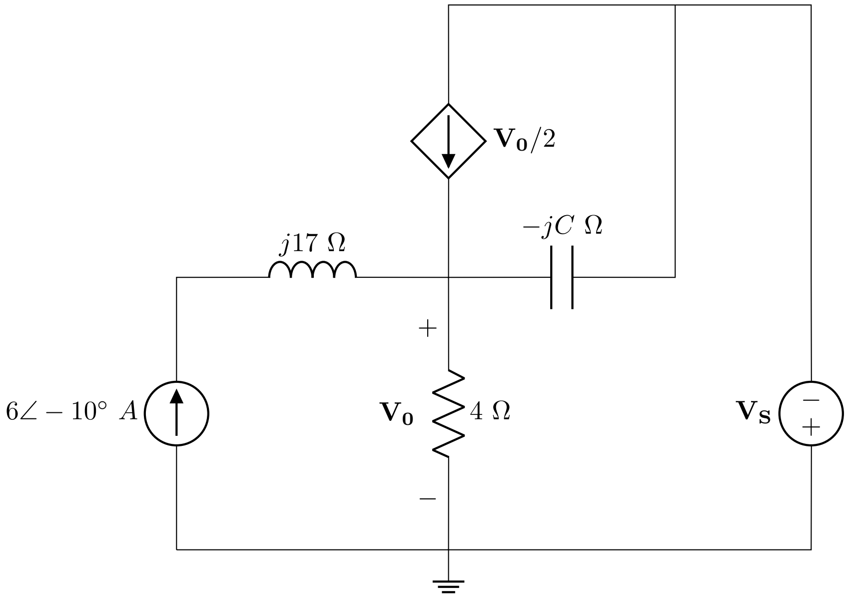
- **Correct**: *1-j2* or *-3+j4*
  - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
  - Capital J is however incorrect, and *1-2j* is also not acceptable.
  - If a complex number has no real part, for example *0+j2*, write *0+j2* as the final answer.
- **Correct**:
  - With angle in degrees: $12\∠34\°$
  - With angle in radians: $12\∠0.5934$
  - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect**:
  - *1-i2* (Use j instead of i)
  - *1 - j2* (no spaces between characters)
  - 12 $\∠$; 34 $\°$; (no spaces between characters)
  - *j10+1* (Real part before the imaginary part for rectangular form.)
  - *-3+4j* (should be $j4$ instead of $4j$)

Questions 1 to 2 [4]	
Use mesh analysis to set up an equation of the following form: $$\alpha \mathbf{I_1} + \beta \mathbf{I_2} = -3$$ Let $R_1 = [R]\ \Omega$, $\mathbf{Z_{C2}} = -j[Z]\ \Omega$ and $\mathbf{V_1} = [V]\angle 45°\ V$. Provide α and β in **rectangular form**.	
	
01	α [2]
02	β [2]

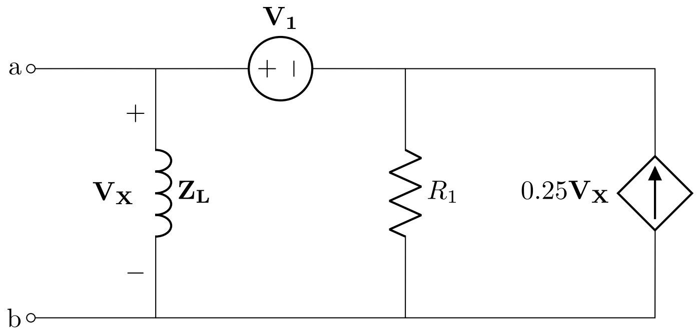
Questions 3 to 6 [4]	
Use nodal analysis to set up two equations of the following form: $$\mathbf{aV_1} + \mathbf{bV_2} = 2$$ $$\mathbf{cV_1} + \mathbf{dV_2} = 8 + j6$$ Give the phasor coefficients **a**, **b**, **c**, and **d** in **rectangular form** when $\mathbf{Z_{C1}} = -j[C]\ \Omega$ and $R_1 = [R]\ \Omega$.	
	
03	**a** [1]
04	**b** [1]
05	**c** [1]
06	**d** [1]

Question 7 [2]	
Use superposition to determine the independent contribution $\mathbf{I_X^A}$ from the current source to $\mathbf{I_X}$. Write your answer in **rectangular form** when $\mathbf{L} = j[L]\ \Omega$, $R_1 = [R]\ \Omega$, $\mathbf{Z_{C1}} = -j[C]\ \Omega$, $\mathbf{V} = 2\angle 0^\circ\ V$, and $\mathbf{I_1} = j[I]\ A$.	
	
07	$\mathbf{I_X^A}$ [2]

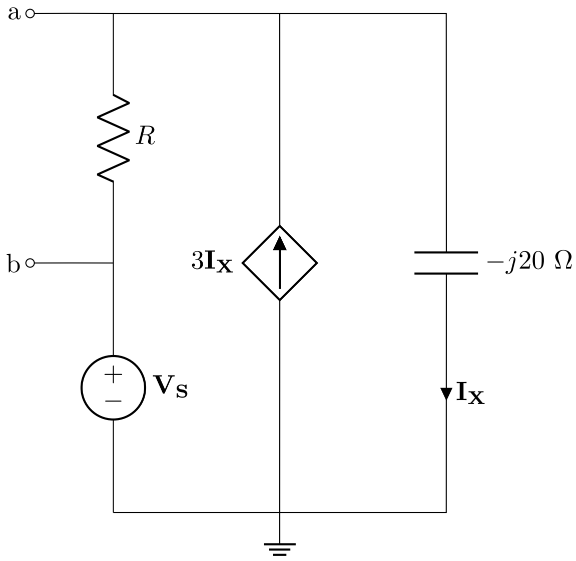
Question 8 [2]	
Use source transformation to determine $\mathbf{V_o}$ as a result of the independent contribution from the voltage source $\mathbf{V_S} = [V]\angle 90°\ V$ if $C = [C]\ \Omega$. Give the answer in **rectangular form**.	



08	$\mathbf{V_0}$ [2]

Questions 9 to 10 [4]	
Let $R_1 = [R]\ \Omega$, $\mathbf{Z_L} = j[L]\ \Omega$, and $\mathbf{V_1} = [V]\ V$. Find the Norton equivalent current $\mathbf{I_N}$ and impedance $\mathbf{Z_N}$ with respect to terminals a-b. Give your answers in **rectangular form**.	
	
09	$\mathbf{I_N}$ [2]
10	$\mathbf{Z_N}$ [2]

Questions 11 to 12 [4]	
Determine the Thevenin equivalent voltage $\mathbf{V_{Th}}$ and impedance $\mathbf{Z_{Th}}$ with respect to terminals a-b. Give your answers in **rectangular form** if $\mathbf{V_S} = [V]\ V$ and $R = [R]\ \Omega$.	



11	$\mathbf{V_{Th}}$ [2]
12	$\mathbf{Z_{Th}}$ [2]

Grand Total : [20]